

MODULE-3**FLIP-FLOPS AND ITS APPLICATIONS**

The Master-Slave Flip-flops (Pulse-Triggered flip-flops): SR flip-flops, JK flip flops, Characteristic equations, Registers, Binary Ripple Counters, Synchronous Binary Counters, Counters based on Shift Registers, Design of Synchronous mod-n Counter using clocked T, JK, D and SR flip-flops.

Combinational Circuit and Sequential Circuit

Key	Combinational Circuit	Sequential Circuit
Definition	A Combinational Circuit is a type of circuit in which the output is independent of time and only relies on the input present at that particular instant.	A Sequential circuit is a type of circuit where output not only relies on the current input but also depends on the previous output.
Feedback	Since output does not depend on the time instant, no feedback is required for its next output generation.	The output relies on its previous feedback so output of previous input is being transferred as feedback used with input for next output generation.
Performance	As the input of current instant is only required in case of Combinational circuit, it is faster and better in performance as compared to that of Sequential circuit.	Sequential circuits are comparatively slower and has low performance as compared to that of Combinational circuit.
Complexity	No implementation of feedback makes the combinational circuit less complex as compared to sequential circuit.	The implementation of feedback makes sequential circuit more complex as compared to combinational circuit.
Elementary Blocks	The elementary building blocks of a combinational circuit are its logic gates.	The building blocks of a sequential circuit are the logic gates along with flip flops.
Operation	Combinational circuits are mainly used for arithmetic as well as Boolean operations.	Sequential circuits are mainly used for storing data.

SEQUENTIAL CIRCUITS- I

Digital logic can be classified into two major sections

1. Combinational logic
2. Sequential logic

Combinational Logic: Combinational logic deals with the technique of “combining” logic gates into circuits that perform some desired function.

Ex: Adders, Subtractors, Decoders, Encoders, Multiplier, Divider.

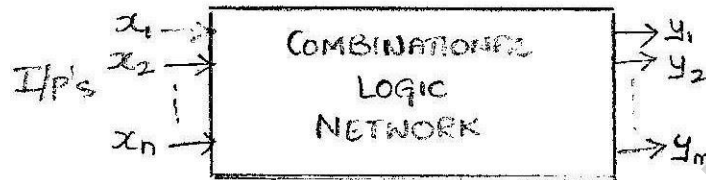


FIG: BLOCK DIAGRAM OF COMBINATIONAL N/W.

- A Combinational network is a two-valued network in which the outputs at any instant of time are **dependent only upon the inputs** present at that instant. Thus the output of a combinational network can be described by a single algebraic expression whose variables are inputs to the network.
- A Combinational network contains no closed loops or feedback paths and hence they are said to be acyclic.
- Combinational circuits are easy to design.
- Truth table is a convenient representation of the operation of a combinational circuits

Sequential Circuits: Many applications in digital world require circuits in which digital outputs are required to be generated in accordance with the sequence in which the input signals are received. Such applications require output to be generated that are not only **dependent on the present input condition but also upon the past history** of these inputs.

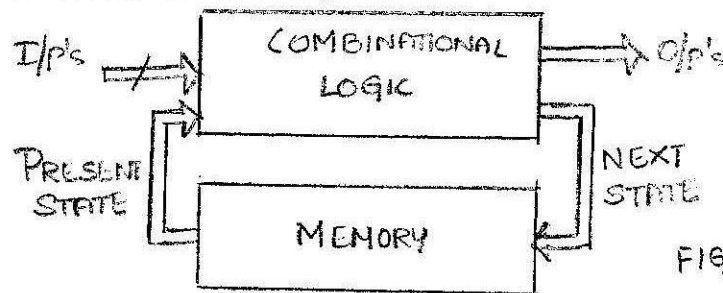


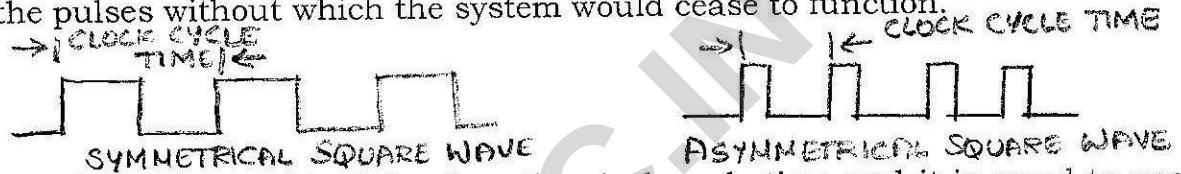
FIG: GENERAL MODEL OF SEQUENTIAL NETWORK

- A Sequential network is defined as a two-valued network in which the outputs at any instant are dependent on only upon the past history or past sequence of inputs.
- The past history of inputs is provided by feedback from output to the input.

- Since the past history of inputs must be preserved by the network, sequential circuits are said to have memory.
- The operation of sequential circuits is expressed in terms of their states. The state or internal state is a set of signals at a given point in the sequential circuit.
- Output of a sequential circuit or network is a function of the present external inputs and its present state i.e. the two determine the next state.
- The basic memory (ability to store information) element in sequential circuits is **Flip flop**.

A Flip flop is a sequential circuit which can store a 1 or a 0 indefinitely. Thus a flip flop is said to possess two stable states. It continues indefinitely in one of these two stable states until a change is prompted by the change in its input.

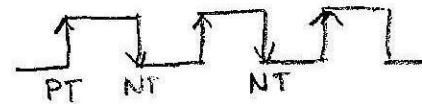
Clock: The heart of every digital system is the system clock. The system clock provides the pulses without which the system would cease to function.



- The basic timing interval is defined as the clock cycle time and it is equal to one period of the clock waveform; it defines the timing interval during which logic operations must be performed.
- There are two basic types of sequential networks.

- Synchronous sequential network:** - In such systems, a change of state occurs in synchronism with the system clock. A change of state will either occur as the clock transitions from low to high or as it transition from high to low.

Low to high transition ----> Positive transition (PT)
High to low transition ----> Negative transition (NT)



A circuit that changes state during PT is called a **positive edge triggered** circuit and that for NT is called as **negative edge triggered** circuit.

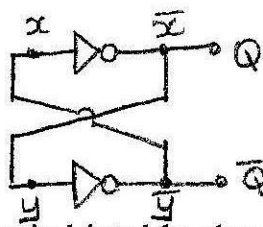
- Asynchronous sequential network:** - There is no synchronization control and hence change in input itself can cause a change in output and state. These networks are capable of high speed performance since it is no longer necessary to wait for the sample times occur before the network responds to the input changes. Since they do not have a synchronizing clock, the memory portion of asynchronous sequential networks consists of unclocked flip flops or time delay elements. A time delay device provides memory by the fact that it takes a finite amount of time for a signal to propagate through it and correspondingly serves to store information for that time duration.

	COMBINATIONAL CIRCUITS	SEQUENTIAL CIRCUITS
1.	Output at any instant of time are dependent only upon the inputs present at that instant	Output at any instant are dependent not only on the present input but also on the past history of these input variables
2.	Memory unit is not required	Memory unit is required to store the past history of inputs
3.	Faster in speed because the delay between input and output is due to propagation delay of gates	Slower in speed
4.	Easy to design Ex: Parallel adder	Comparatively not easier Ex: Serial adder

	SYNCHRONOUS SEQUENTIAL CIRCUITS	ASYNCHRONOUS SEQUENTIAL CIRCUITS
1.	Memory elements are clocked flip flops	Memory elements are either unclocked flip flops or time delay elements.
2.	Change in input signals can cause a change in output or state only on activation of clock signal.	Change in input itself can cause change in output or state.
3.	Maximum operating speed of clock depends on time delays involved.	Due to absence of clock they can operate faster than synchronous circuits
4.	Easier to design	Difficult to design

Propagation time delay is the time required for a signal to pass from the input of a circuit to its output.

Basic Bistable Element:



- Central to all flip flops is the basic bistable element as shown above. It consists of two cross-coupled not gates i.e. output of first not gate serves as input to the second not gate & output of second serves as input to the first not gate
- A basic bistable element is a circuit having two stable states
 Let $x=0$ then $Q = \bar{x} = 1$
 So $y=1$ then $\bar{Q} = \bar{y} = 0$
 Thus $\bar{Q} = x = \bar{y} = 0$ & $Q = \bar{x} = y = 1$ (circuit is stable with these values)
 Similarly let $x=1$ then $Q = \bar{x} = 0$

So $y=0$ then $\bar{Q} = \bar{y} = 1$

Thus $\bar{Q} = x = \bar{y} = 1$ & $Q = \bar{x} = y = 0$ (2nd stable state)

- Since it has two stable states, the basic bistable element is used to store binary symbols. In case of positive logic, when $Q=1$ the element is said to store a 1 and when $Q=0$, the element is said to store a 0.
- The binary symbol that is stored in the basic bistable element is referred to as the content or state of the element. The state is given by the signal value at the Q output terminal. Thus Q output is called normal output & $\bar{Q}$ is referred to as Complementary output.
- When the device is storing a 1 it is said to be in **1-state or set state** and when the device is storing a 0 it is said to be in **0-state or reset state**.
- A Third equilibrium state can exist when the two output signals are about halfway between logic-0 and logic-1. Output is not a valid logic signal. This is known as the metastable state.
- When power is applied, it becomes stable in one of its two stable states. It remains in this state until power is removed.

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- **A flip flop** is a bistable device, with inputs, that remain in a given state as long as power is applied and until input signals are applied to cause change in its output.

The process of storing a 1 into a flip flop is called **setting or presetting** the flip flop; while the process of storing a 0 into a flip flop is called **resetting or clearing** the flip flop

- Inputs to a flip flop are two types
Asynchronous input
Synchronous input

Asynchronous input: is one in which a signal change of sufficient magnitude and the duration essentially produces an immediate change in the state of the flip flop.

Synchronous input: is one that does not immediately affect the state of the flip flop but rather affects the state of the flip flop only when some control signal, usually called as enable or clock input also occurs.

Latches: - Latches and flip flops are the basic building blocks of the most sequential circuits; the difference is in the method used to change their states.

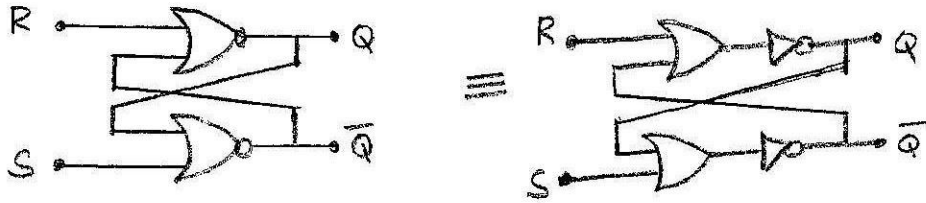
- Flip flop changes state only at times determined by the clocking state.
- Latches changes state corresponding to changes in input independent of the clocking signal. A special control signal called as enable or clock is provided with the latch, when enable signal is active output changes occur as input changes.

- But when enable signal is not activated input changes do not affect output, such latches are called as **gated latch**.

1. **SR Latch (Set-Reset):-**

- Simplest type of latch that consists of two cross-coupled NOR gates

LOGIC
DIAGRAM



- It has two inputs namely S (set) and R (Reset) and two outputs Q & $\bar{Q}$.

FUNCTION
TABLE

INPUTS		OUTPUTS	
S	R	Q^+	$\bar{Q}^+$
0	0	Q	$\bar{Q}$
0	1	0	1
1	0	1	0
1	1	0*	0*

* UNPREDICTABLE STATE

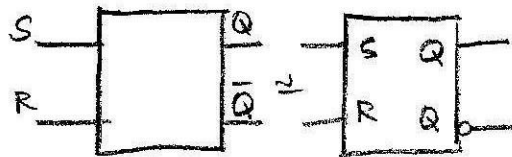
- **Case 1:** When $S = R = 0$, Circuits becomes similar to basic bistable element i.e., cross-coupling of two NOT-gates thus latch is in one of the two stable states. In the Truth table Q & $\bar{Q}$ represents the state of the latch when the inputs are applied i.e. Q is the present state, Q^+ & $\bar{Q}^+$ represents the response of the latch to the inputs applied. i.e. $\bar{Q}^+$ is the next state of the latch.

Thus the outputs do not change and the next state equals the present state.

$$\text{i.e. } Q^+ = Q$$

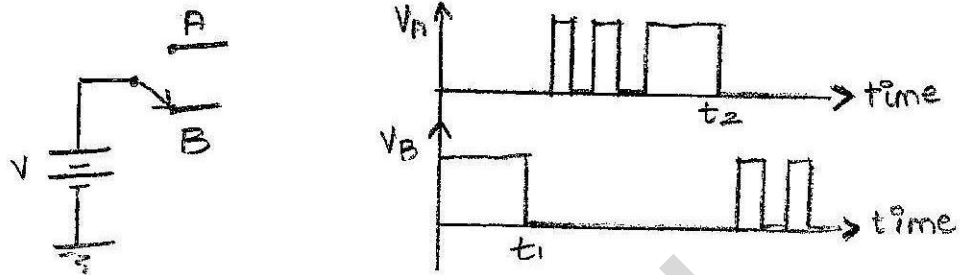
- **Case 2:** When $S=0, R=1$, Regardless of the second input to the upper NOR gate, the output Q must become 0 since $R=1$. This in turn is fed to the lower NOR gate. Thus $Q=0$ & $\bar{Q}=1$. (reset state)
- **Case 3:** When $S=1, R=0$, The latch becomes set regardless of its present state because one of the input to the lower NOR gate is a 1. Thus $Q=1$ & $\bar{Q}=0$. (set state)
- **Case 4:** When $S=1, R=1$, Outputs of both NOR gates becomes '0' and hence it is a forbidden state. When both S & R transit to zero simultaneously, then the latch could get into the metastable state which is undesirable.

SYMBOL



An application of SR latch: A switch debouncer

- When mechanical switches such as toggle switches or push buttons are switched from one position to the other, several make and break operations occur at the second position called **Switch Bounce**.



- At time t_1 , the centre contact is moved from B to A first the voltage at B becomes zero as soon as contact leaves B. However, when it reaches A, several make and break operations takes place due to the spring like nature of the contacts.

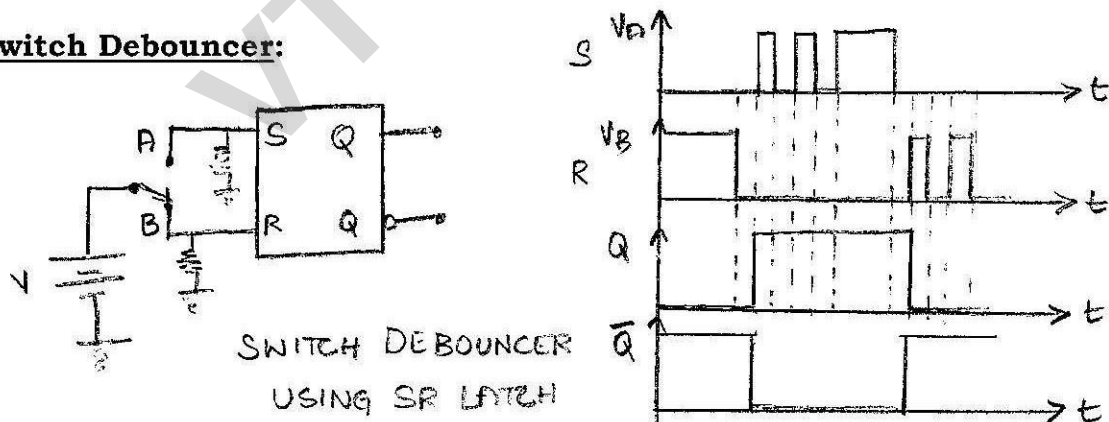
➤ During contact bounce, the center contact of the switch does not return all the way to terminal B. similar occurrence can be seen at B, when center contact is moved from A to B at t_2 . This effect is called **switch bounce** and is normally undesirable.

Ex: Push-button keys on a keyboard, contact bounce causes a system to respond as through a key was pressed several times in succession.

➤ This effect is undesirable even in digital circuits because between t_1 & t_2 , the logic level read by a gate connected to A could be a 0 or a 1 depending on the instance when the switch is read, leading to unpredictable results.

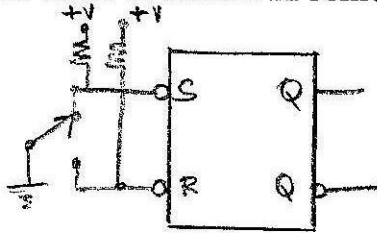
Such a switch bounce can be eliminated by the use of a SR latch.

A Switch Debouncer:



- Let center contact of the switch be initially at position B, i.e. $S=0$, $R=1$
Therefore $Q=0$, $\bar{Q}=1$ for time $t < t_1$
- At time t_1 , switch moves from position B to A, after the center contact leaves B till it reaches A, $S = R = 0$, thus output remains $Q=0$, $\bar{Q}=1$
- As soon as contact ouches A, he inputs to the latch become $S=1$, $R=0$, therefore $Q=1$, $\bar{Q}=0$

- Now, when the center contact temporarily leaves A due to contact bounce, the inputs becomes $S=R=0$, thereby Q remains 1 & $\bar{Q}=0$. Similar action takes place when center contact is switched back to B at $t = t_2$
- Thus SR latch is useful in removing the effect of contact bounce as glitches.



NOTE: WAVEFORMS REMAIN SAME AS OF SR LATCH SWITCH DEBOUNCER

FIG: SWITCH DEBOUNCER USING $\bar{S}\bar{R}$ LATCH

$\bar{S}\bar{R}$ Latch:

- The $\bar{S}\bar{R}$ latch is constructed by cross-coupling two NAND gates

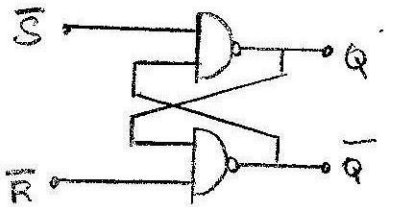


FIG: LOGIC DIAGRAM

INPUTS		OUTPUTS	
$\bar{S}$	$\bar{R}$	Q^+	$\bar{Q}^+$
0	0	1*	1*
0	1	1	0
1	0	0	1
1	1	Q	$\bar{Q}$

FORBIDDEN STATE

FIG: FUNCTION TABLE

- It has two inputs, namely $\bar{S}$ & $\bar{R}$ and two outputs Q & $\bar{Q}$
- When $\bar{S} = 1$; $\bar{R} = 1$, Circuits becomes similar to basic bistable element i.e. Cross-coupling of two NOT gates. Thus the device has 2 stable states.
- When $\bar{S} = 0$, $\bar{R} = 1$, (set state) The latch becomes set regardless of the second input to the upper NAND gate, therefore $S = 0$ thus $Q=1$ & $\bar{Q}=0$
- When $\bar{S} = 1$ $\bar{R} = 0$, (reset state) The latch becomes reset and hence $Q = 0$ & $\bar{Q} = 1$
- When $\bar{S} = \bar{R} = 0$, (forbidden state) Outputs of both NAND gates becomes '1' and hence forbidden state
 - If both S & R transit to 1 simultaneously, then the latch could enter into the metastable state which is undesirable.

Ex: Switch debouncer using $\bar{S}\bar{R}$ latch

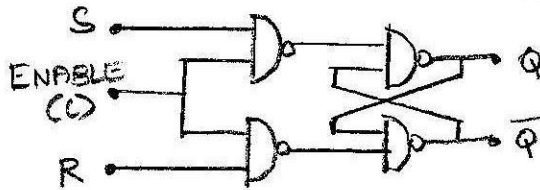
LOGIC DIAGRAM AND WAVEFORMS AS GIVEN ABOVE

FIG: SYMBOL OF $\bar{S}\bar{R}$ LATCH



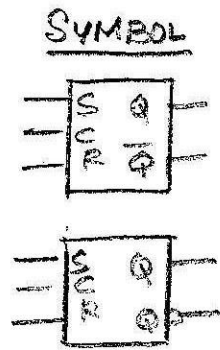
Gated SR Latch

- Gated SR latch is also called as **SR latch with enable**.



LOGIC DIAGRAM

S	R	C	Q^+	$\bar{Q}^+$
0	0	1	Q	$\bar{Q}$
0	1	1	0	1
1	0	1	1	0
1	1	1	1*	1*
x	x	0	Q	$\bar{Q}$



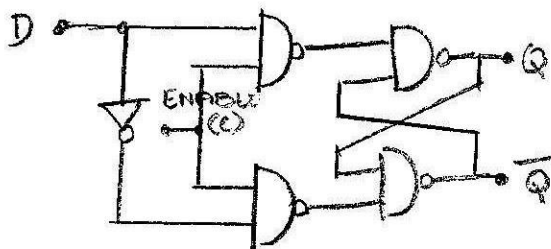
- A gated SR latch is as shown above. It consists of the $\bar{S} \bar{R}$ latch along with two additional NAND gates and a control input, C referred to as the enable, gate or clock input.

The enable input decides when the inputs S & R are effective i.e. it decides if a change in input causes corresponding changes in output of flip flop

- As long as the enable input is zero, the output of the two NAND gates (X & Y) is 1. This output is in turn fed as input to the NAND gates (A & B) and hence according to the function table of $\bar{S} \bar{R}$ latch, $\bar{S} \bar{R}$ latch remains in its current stable state. The latch is set to be disabled.
- When the enable input is made high, the latch is said to be enabled.
- The NAND gate inverts the signals on the S and R input lines. When the latch is enabled.
- Since the effects of the S and R inputs are dependent upon the presence of an enable signal, these inputs are classified as synchronous inputs.

Gated D Latch (Data Latch)

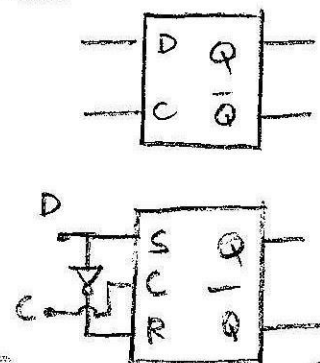
- SR, $\bar{S} \bar{R}$ & gated $\bar{S} \bar{R}$ latch had an inputs combination that was forbidden, in particular when the inputs S & R or $\bar{S}$ & $\bar{R}$ are simultaneously high. Hence a gated D-latch was designed to overcome this problem.
- The logic diagram of a gated D-latch is as shown



LOGIC DIAGRAM

D	C	Q^+	$\bar{Q}^+$
0	1	0	1
1	1	1	0
x	0	Q	$\bar{Q}$

FUNCTION TABLE



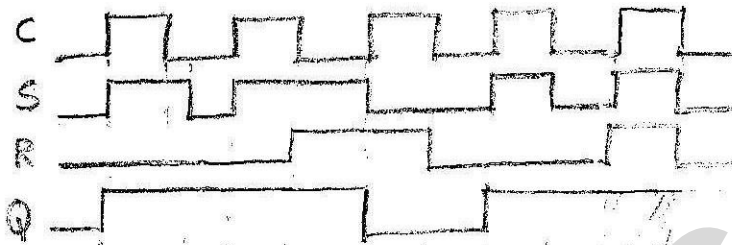
SYMBOL

- A NOT gate is connected between the S and R terminals thus the latch consists of a single input 'D' that determines the next state. It also consists of a control signal, i.e. enable or clock that determines when the 'D' input is effective.

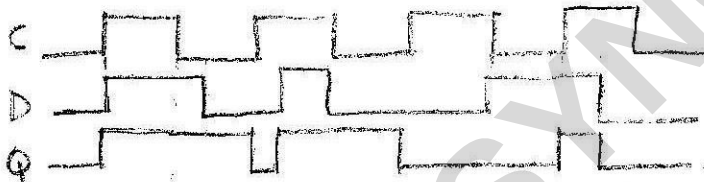
- When the latch is enable $C=1$, the output of the latch follows the D-input. i.e. if D is 0, then output switches to or remains in the 0-state and vice-versa.
- If $C=0$ i.e. the latch is disabled, then the latch remains in the previous state itself.
- D-flip flop (Transparent latch):- A flip flop whose output follows its data input when the clock is active.

Timing diagram is a graph that depicts the input and output transactions of a flip-flop as a function of a time.

Timing diagram for an SR latch [GATED]



Timing diagram for an gated D latch



- Applying a '1' simultaneously to both the S and R input terminals when $C=1$ should be avoided to prevent the possibility of an unpredictable state when the signals are removed subsequently or if C is changed to 0 while both signals are '1'.

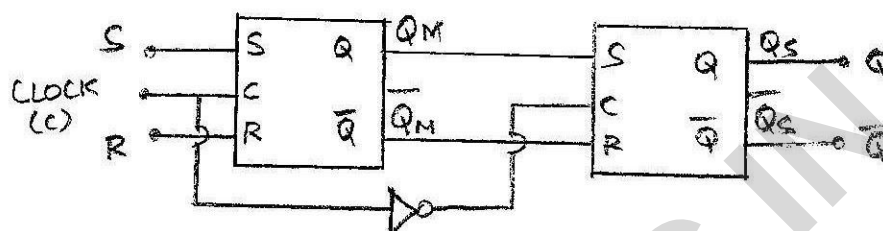
Master-Slave Flip Flops (Pulse Triggered Flip Flop):

- Flip flops can be categorized
 - Pulse triggered flip flops
 - Edge triggered flip flops
- Latches respond immediately to changes in input. The enable or clock i.e. the control signal might be present wherein changes in input cause changes in output only when the enable is high. This property is referred as **transparency**.
- In some applications, this property is undesirable wherein it is necessary that the output changes occur only with changes in a control input line like the clock.
- The purpose of master-slave flip flops is to protect the flip flop's output from changing or responding to glitches on the input. Master-slave flip flops are used in applications where glitches are prevalent on inputs.

- A master-slave flip flop consists of two cascaded sections, 1st section is referred to as master and 2nd section as the slave.
 - Information is entered into the master on one edge or level of the control signal and the information is transferred to the slave on the next edge or level of the control signal.
- Each section is a Latch.

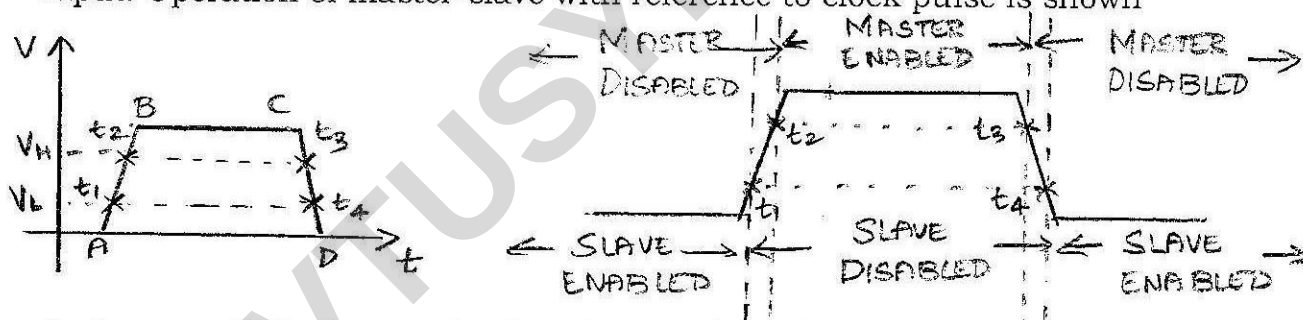
Master Slave SR Flip Flops:

- The master – slave SR flip-flop is constructed using two gated SR latches and an inverter as shown below



MASTER - SLAVE SR FLIP FLOP

- The S & R inputs are used to set and reset the flip-flop. The clock acts as a control input. Operation of master-slave with reference to clock pulse is shown



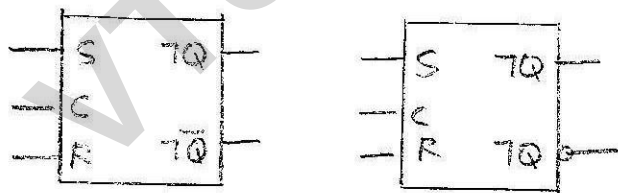
- As long as $C=0$, the master i.e. the gated SR latch is disabled and hence any changes on the S & R input lines are ignored. But the slave is enabled due to the presence of an inverter.
- Since the slave is enabled, it is in the same state as that of the master because the outputs of the master Q_M & $\bar{Q}_M$ are connected to S & R inputs respectively, i.e. State of master is transferred to the slave when clock is at logic 0
- At time t_1 , the slave is disabled but retains the state of master.
- The control signal continues to rise and at time t_2 , the master is enabled and it responds to the inputs S & R.
- Changes in the state of the master are not reflected to the slave since the slave is disabled.
- At time t_3 , the control signal is returned to its low level and hence the master gets disabled.
- At time t_4 , the slave is enabled and hence it takes the state of the master.

- For very short instants of time, during rising and falling edges of the control signal both the master and the slave latches are disabled.
- The master can change its state anytime when the clock input is logic 1, whereas the slave can change state only along the falling edge of the clock input at time t_4 .
- Thus, it can be observed that the output change of the master-slave flip flop is synchronized to the falling edge of the control signal.

Function table of master-slave SR-flip flop:

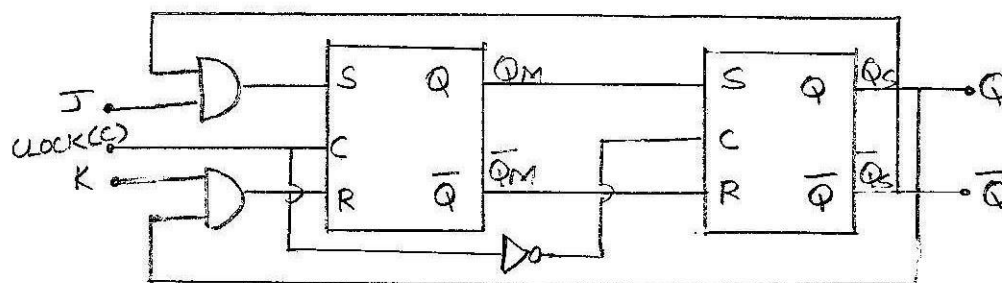
S	R	C	Q^+	$\overline{Q}^+$
0	0	1	Q	$\overline{Q}$
0	1	1	0	1
1	0	1	1	0
1	1	1	Undefined	
X	X	0	Q	$\overline{Q}$

- The clock pulse shown in the table represent the fact that the master is enabled as long as the clock is at 1 and the state of the master is transferred to the slave when the clock begins to go to '0'.
- When $S=R=1$, when control signals goes from high to low. The master enters into an unpredictable state; this state is transferred to the slave. Thus, the output of the master-slave flip flops itself becomes unpredictable. Hence such a condition is undesirable and should be avoided.
- Since, the state of the master depends on the period of time for which the clock is at logic 1 and the state of the slave changes at the falling edge these flip flops are also called as pulse triggered flip flops.
- The two logic symbols of the master-slave SR flip flop



- The '⌋' symbol represents the fact that the output changes only at the falling edge of the pulse and is called as the postponed output indicator. Since the output is postponed till the end of the pulse period.

Master-Slave JK Flip Flop(MS- JK)



MASTER - SLAVE JK FLIP FLOP

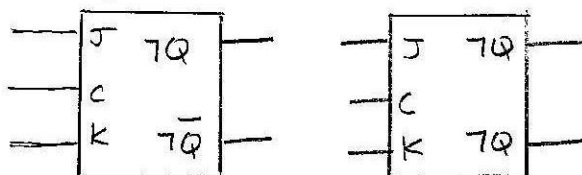
- A master slave JK flip flop is constructed using a MS SR flip flop and two AND gates as shown above.
- The main purpose of a MS JK flip flop is to overcome the undefined output state of MS SR flip flop when $S=R=1$ during ON pulse of the clock.
- It has two inputs namely, J & K that are analogous to S & R inputs. Thus J & K inputs are used to set and reset the flip flop respectively.
- The truth table of a master-slave JK flip flop is as shown

J	K	C	Q ⁿ	Q ⁿ⁺¹
x	x	0	Q	Q
0	0	1	Q	Q
0	1	1	Q	0
1	0	1	Q	1
1	1	1	Q	Q-bar

- Consider the case when $J=K=1$. Let $Q=0$ & $\bar{Q}=1$, thus the output of the lower, i.e. K-input AND gate is 0 and the output of the J-input AND gate is a 1, as a result of feedback. Thus $R=0$, $S=1$ to the master flipflop.
- At the instant the clock goes high, master sets i.e. $Q_M=1$ and $\bar{Q}_M=0$.
- On the next falling edge of the clock, the state of the master is transferred to the slave thus $Q_S=1$ & $\bar{Q}_S=0$.
- Similar condition can be observed when $Q=1$ & $\bar{Q}=0$ wherein the output changes to $Q_S=0$ & $\bar{Q}_S=1$.
- When $J=K=0$, both the AND gate outputs are zero and this corresponds to $S=R=0$. Thus the next state is the same as the present state.
- Race - around toggling condition of JK FF when $J=K=1$.

‘ $\neg$ ’ -> postponed output indicator

SYMBOLS

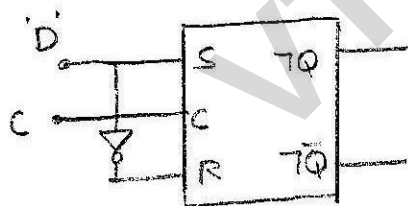


0's catching and 1's catching:

- A Master-slave JK flip flop is subject to 0's & 1's catching
- If the slave latch is in its 1-state then if first IP's to flip flop are $J=0$, $K=1$ causes master to reset.
The slave resets during falling edge of the clock.
- Once the master is reset, any changes in J & K inputs are not reflected by master since the slave does not change its state until C returns to 0 (negative edge)and the feedback from slave keeps master in the same state. This is 0's catching; similarly we have 1's catching.

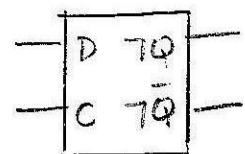
Additional type of Master - Slave flip flops:

MASTER-SLAVE 'D' FF: - By placing an inverter between the S & R inputs, of a MS SR FF a MSD-FF can be realized.



FUNCTION TABLE

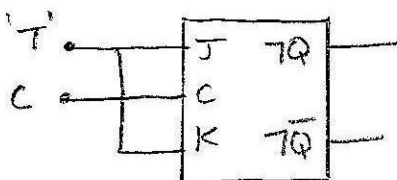
D	C	Q^+	$\bar{Q}^+$
X	0	Q	$\bar{Q}$
0	1	0	1
1	1	1	0



SYMBOL

MASTER - SLAVE 'T' FF: - By shorting the inputs J&K of a MS JK FF we get a MS T-FF.

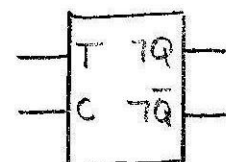
- The flip flop toggles when $T=1$ & does not change its state when $T=0$



LOGIC DIAGRAM

T	C	Q^+	$\bar{Q}^+$
X	0	Q	$\bar{Q}$
0	1	Q	$\bar{Q}$
1	1	$\bar{Q}$	Q

FUNCTION TABLE



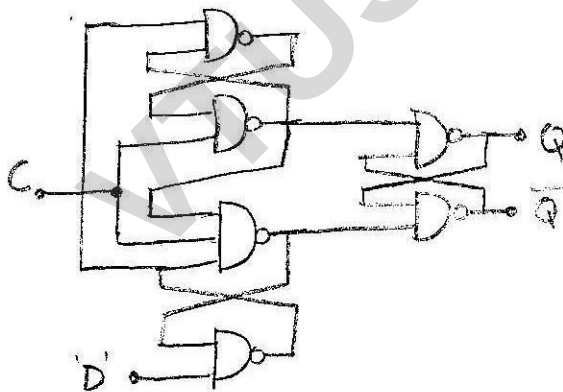
SYMBOL

Edge triggered flip flops:

- In a master slave flip flop, the master is enabled as long as clock is 1, this leads to 0's catching & 1's catching
- To avoid the 0's & 1's catching problem, it should be seen that the J & K inputs are held constant when the clock input is 1, i.e. the master is enabled.
- In this way, the state of the master can be established during the positive edge of the clock and then transferred to the slave during the negative edge of the clock.
- Such flip flops are called as edge triggered flip flops. These flip flops respond only at the edges of the clock and ignore any changes in the inputs in between triggering edges
- Edge-triggered flip flops can be classified as
 - Positive edge triggered flip flops
 - Negative edge triggered flip flops

Positive edge triggered D flip flop:

- Logic diagram of positive edge triggered D flip flop is as shown below.
- D is the input & C is the control or the clock input.



LOGIC DIAGRAM

D	C	Q ⁺	Q ⁺ ⁺
0	↑	0	1
1	↑	1	0
X	0	Q	Q ⁺
X	1	Q	Q

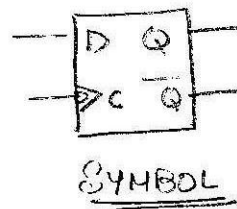
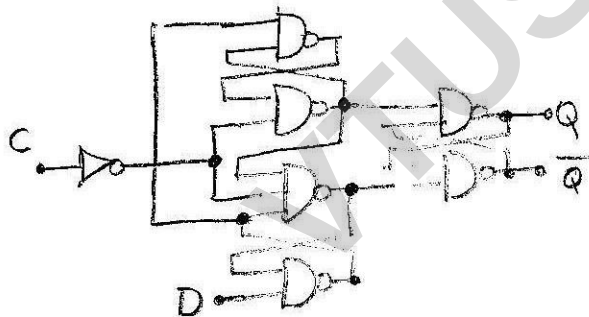
SYMBOL

- Positive edge-triggered means the setting or resetting of the flip flop is established during the rising or positive edge of the clock.
- The '↑' symbol in the column indicates that D appears at the output during the positive edge of the clock.
- Let C=0, thus regardless of the 'D' input, the output of NAND gates 2 & 3 is a '1', thus input to the $\bar{S} \bar{R}$ latch is $\bar{S}=1=\bar{R}$ and the output remains in the same state
- If D=0, output of gate 4 is '1' & output of gate 1=0

- Let the clock, now transit from 0 to 1, i.e. during the positive edge of the clock, all the three inputs to gate 3 becomes zero and hence output of gate 3 is 0, the output of gate 2 remains a 1.
Now the input to $\bar{S} \bar{R}$ latch is $\bar{S}=1$ & $\bar{R}=0$, thus $Q=0$ & $\bar{Q}=1$
- Thus during positive edge of the clock the 'D' input is transferred to the output.
- Also the output of the gate 3 is fed back to the input of gate 4. This makes the output of the gate 4 to be '1' and subsequent changes in 'D' input will have no effect.
- Thus, after the occurrence of the positive edge of the clock when $D=0$ the flip flop is in its 0-state and any changes in 'D' input are not allowed even though clock is 1
- Hence, only on the occurrence of the positive edge of the clock signal does the flip flop respond to 'D' input and changes in 'D' when $C=1$ is ineffective.

Negative-Edge triggered flip flops:

- Here the occurrence of the negative edge of clock signal causes the output changes with respect to changes in input.
 - It can be realized by placing an inverter at the control input of the flip flop
- Logic diagram of positive edge triggered D-flip flop



D	C	Q ⁺	Q ⁺
0	↓	0	1
1	↓	1	0
X	0	Q	Q
X	1	Q	Q

FUNCTION TABLE

(EXPLAIN ACCORDINGLY)

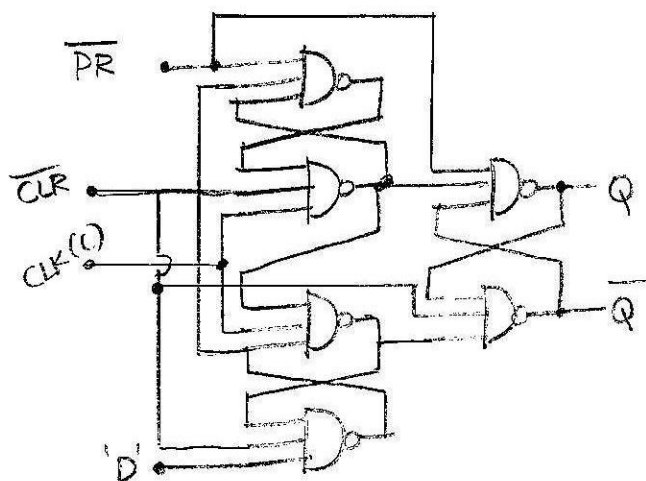
Asynchronous Inputs:

- To provide greater flexibility many flip flops have both synchronous and asynchronous inputs.

The two main asynchronous inputs are

- **PRESET (PR)**
- **CLEAR (CLR)**

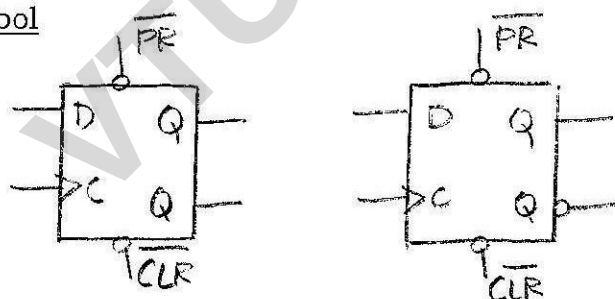
- The asynchronous inputs $\overline{PR}$ & $\overline{CLR}$ are used to forcibly set and reset the flip flop respectively independent of the control input.
- The logic diagram of a positive edge triggered D-flip flop with asynchronous preset & clear inputs is shown



$\overline{PR}$	$\overline{CLR}$	D	C	Q^+	$\overline{Q}^+$
0	1	x	x	1	0
1	0	x	x	0	1
0	0	x	x	1*	1*
1	1	0	↑	0	1
1	1	1	↑	1	0
1	1	x	0	Q	$\overline{Q}$
1	1	x	1	Q	$\overline{Q}$

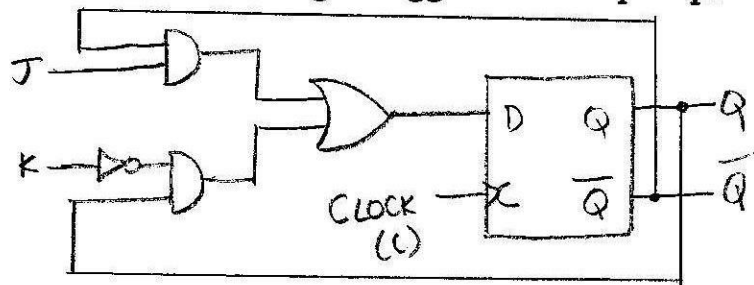
- A logic-0 on the $\overline{PR}$ input line causes the flip flop to enter into 1-state (set) asynchronously
- A logic-0 on the $\overline{CLR}$ input line causes the flip flop to enter into 0-state (reset) asynchronously.
- If $\overline{PR}=0$ & $\overline{CLR}=1$ while $C=0$, then output of NAND gate becomes 1 & gate 6 becomes 0. Thus $\overline{S}$ $\overline{R}$ latch is forced to set, similarly if $\overline{PR}=1$ & $\overline{CLR}=0$, the $\overline{S}$ $\overline{R}$ latch portion of flip flop is forced to be reset.

Logic symbol

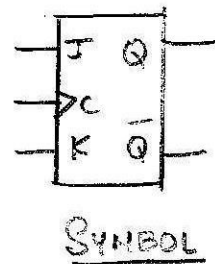


Additional types of edge triggered flip flops

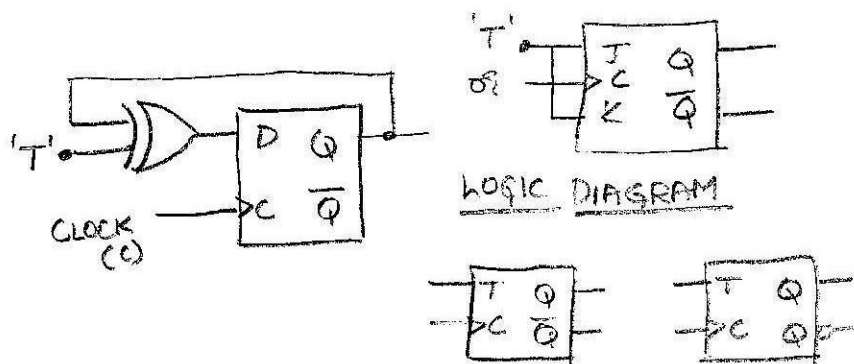
1. Positive Edge Triggered JK flip flops



J	K	C	Q^+	$\overline{Q}^+$
0	0	↑	Q	$\overline{Q}$
0	1	↑	0	1
1	0	↑	1	0
1	1	↑	$\overline{Q}$	Q
x	x	0	Q	$\overline{Q}$
x	x	1	Q	$\overline{Q}$



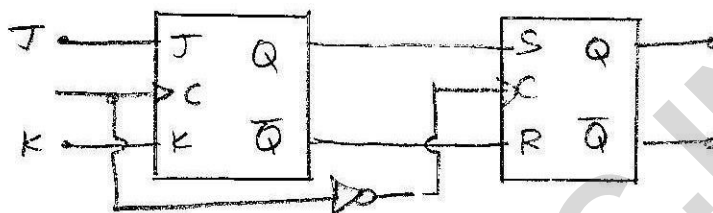
2. Positive Edge Triggered T flip flop



T	C	Q ⁺	Q ⁺ ₊
0	↑	Q	Q
1	↑	Q	Q
X	0	Q	Q
X	1	Q	Q

FUNCTION TABLE

Master-Slave flip flop with Data lockout:



- To have delayed outputs from flip flops, a master slave flip flop is appropriate. But it is subjected to 0's & 1's catching.
- However, to avoid 0's & 1's catching; the master should respond to the information lines only on one edge of the control signal and to transfer to the slave on the next opposite edge.
- Master – slave flip-flops having this property are said to have data-locked.
- In the above diagram, master is a positive edge triggered JK flip flop and SR latch is used for the slave.
- Here, information is entered into the master during the positive edge of the clock and that is transferred to the slave during the negative edge.
- This flip-flop is not subjected to 0's and 1's catching.

SEQUENTIAL CIRCUITS- II

Characteristic equations:

- A variation in the function table gives the next state table. The next state table shows the value of the next state of the flip flops for each combination of values to the present state of the flip flop.
- The algebraic description of the next state table of a flip flop is called as the characteristic equation of the flip flop and is obtained by constructing a K-Map for Q^+ .

Characteristic equation of an SR flip-flop:

Function table of S-R flip flop

S	R	Q^+
0	0	Q
0	1	0
1	0	1
1	1	X

Next state table of S-R flip flop

S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

K-Map

S	RQ			
	00	01	11	10
0	0	0	0	0
1	1	1	X	X

$$Q^+ = S + \overline{R}Q$$

Characteristic equation of a JK flip-flop:

Function table of J-K flip flop

J	K	Q^+
0	0	Q
0	1	0
1	0	1
1	1	$\overline{Q}$

Next state table of J-K flip flop

J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

K-Map

J	KQ			
	00	01	11	10
0	0	0	0	0
1	1	1	0	1

$$Q^+ = J\overline{Q} + \overline{K}Q$$

Characteristic equation of a D flip-flop:

Function table of D flip flop

D	Q^+
0	0
1	1

Next state table of D flip flop

D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

K-Map

D	Q	
	0	1
0	0	0
1	1	1

$$Q^+ = D$$

Characteristic equation of a T flip-flop:

Function table of D flip flop

T	Q^+
0	Q
1	$\overline{Q}$

Next state table of D flip flop

T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

K-Map

T	Q	
	0	1
0	0	1
1	1	0

$$Q^+ = T \oplus Q$$

REGISTERS:

- Registers and counters are important applications of clocked flip flops.
- A collection of flip flop in cascade is called as a REGISTER.
- Registers are used to store data in a digital system so as to make it available to the logic elements during the computing process.
- A register can consist of only a finite number of flip flops and each flip flop can store either a '0' or a '1'. Each combination of 1's and 0's stored in a register is referred to as the state or content.

Thus a cascade of 4 flip flops configured as a register can store 4 bits, i.e. a nibble of data. A 4-bit register can store binary bits from '0000' to '1111'. These are called states. Thus a 4-bit register has 16 possible states.

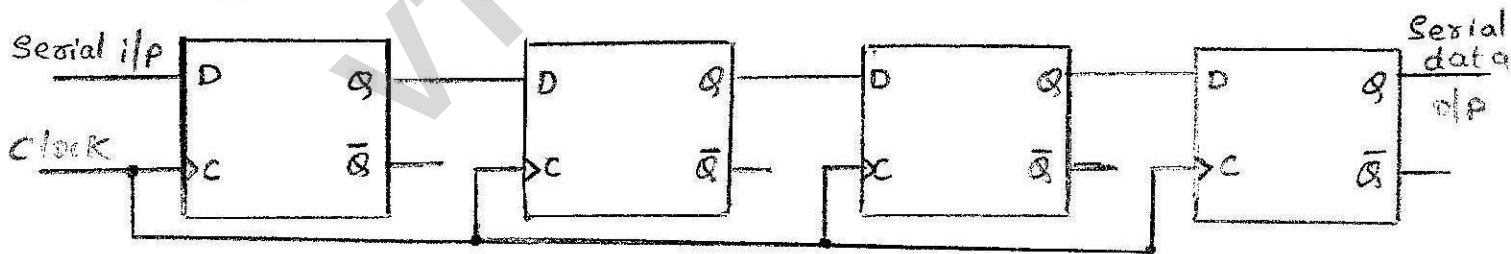
- Registers that are capable of moving data stored in their flip flops is referred to as a **shift-register**.
- Shift registers that can shift data in both the directions are called as **Bi-directional shift register** and the shift registers that can shift data in only one direction are called as **unidirectional shift register**.
- Shift registers can also be classified based on whether information is entered into and taken out from a register either serially or parallel.

Accordingly there are four possible registers:

- Serial in Serial Out (SISO)
- Serial in Parallel Out (SIPO)
- Parallel in Parallel Out (PIPO)
- Parallel in Serial Out (PISO)

4- Bit Serial In Serial Out Unidirectional Register:

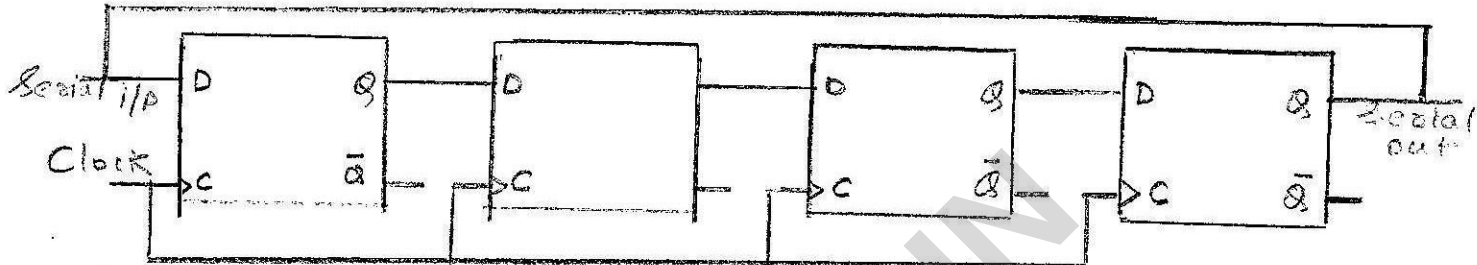
4-bit Serial in Serial out (SISO) unidirectional shift register configured using **positive edge triggered D-flip flop** is as shown.



- The Q output of each flip flop is connected to the D-input of the next flip flop. The control input, i.e. the clock input of each flip flop is connected together to a common synchronizing signal called the clock. Thus, on the occurrence of the positive edge of the clock signal, the content of each flip flop is shifted one position to the right.
- The output from the shift register occurs at the rightmost flip flop on the serial-data-out line.
- The serial data input applied, appears at the serial data output after 4 clock pulses since it consists of 4- flip flops.

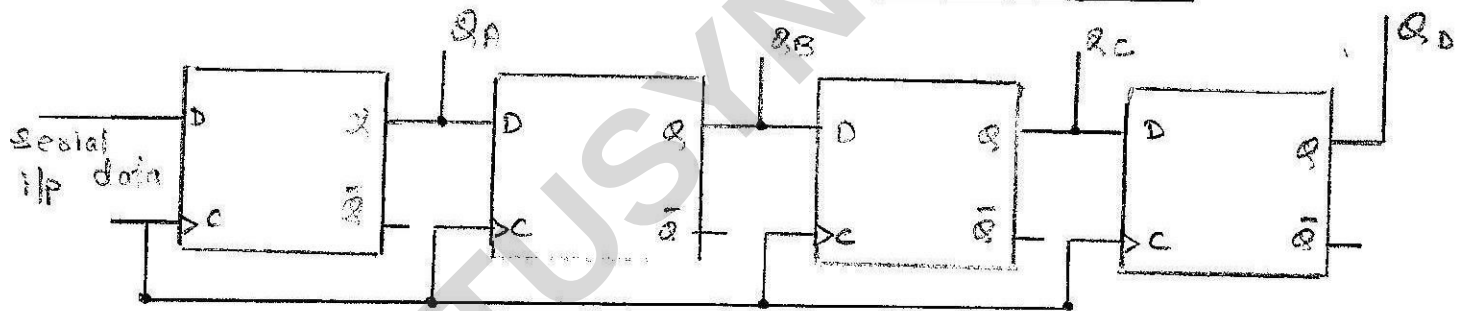
- Ex: assume a logic 1 to be applied at serial data in line and before the positive edge of the clock let the content of the register be 0011 thus after the positive edge of the clock signal, the register content is 1001.
- The content of the last flip flop is lost after every clock pulse.
- To store the information within a register, the serial-data out line is connected to the serial-data in line. Such registers are called as **Circular Shift Registers**.

4-bit Circular Shift Register



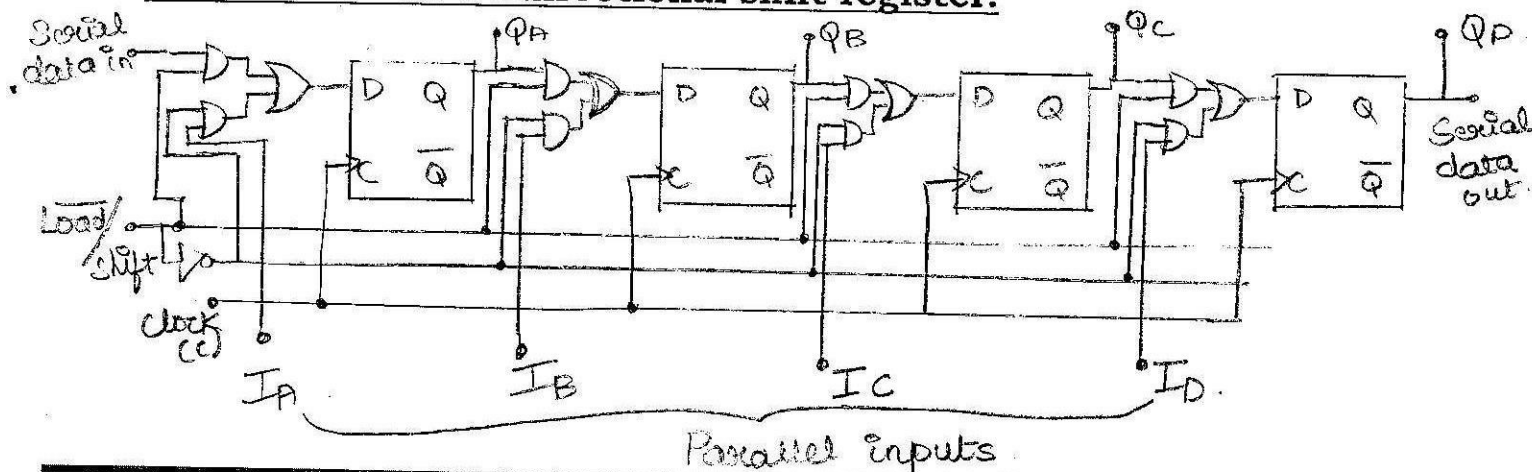
- Thus a 1001 stored in the register becomes 1100 on one shift to the right caused by the application of one clock pulse.

4-Bit Serial In Parallel Out Unidirectional Shift Register:



- A 4-bit SIPO unidirectional shift register is as shown above. Outputs are obtained from each flip-flop.
- Data is entered into the register serially and the output is available as a single entity i.e. parallel output at the flip flop output terminal.

4-bit Parallel-In unidirectional shift register:

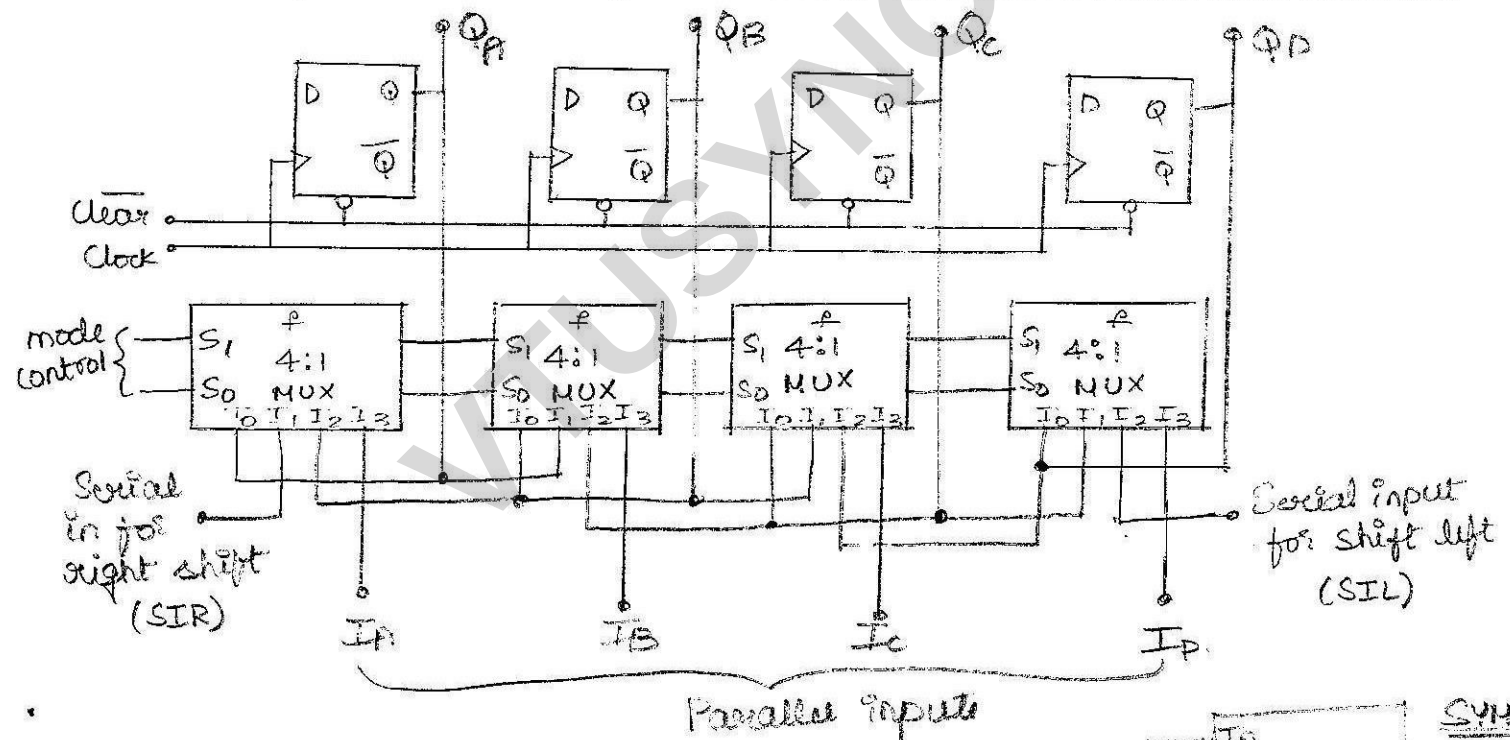


- The Parallel-In unidirectional shift register is as shown above. The operation of the register is controlled by the $\overline{\text{LOAD}}$ / **SHIFT** line.
- When the $\overline{\text{LOAD}}$ / SHIFT line is at logic '0', the lower AND gates are enabled and the signal on the parallel data input lines I_A, I_B, I_C, I_D are transferred into the register upon the occurrence of a positive edge of the clock.
- When a logic '1' is applied to the $\overline{\text{LOAD}}$ / SHIFT line, the upper AND gates are enabled and it works as a unidirectional shift register.
- This register can also be configured as SISO and SIPO.

BIDIRECTIONAL SHIFT REGISTER:

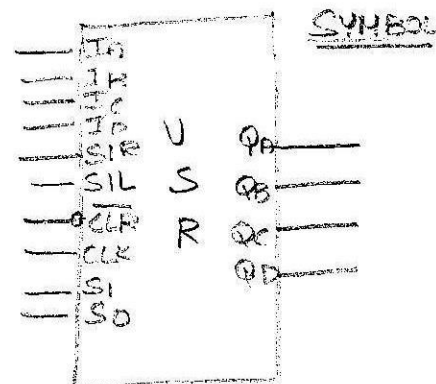
UNIVERSAL SHIFT REGISTER:

- A universal shift register is one which is capable of shifting its contents in both directions either left or right depending upon the signals on the control lines.
- A universal shift register also has the capability to accept both serial and parallel inputs as well as capability of serial and parallel output.
- The operation of a shift register is based on the **mode selection table** as follows:



MODE SELECTION TABLE

S_1	S_0	Data line selected	Operation
0	0	I_0	Hold
0	1	I_1	shift right
1	0	I_2	shift left
1	1	I_3	Parallel load

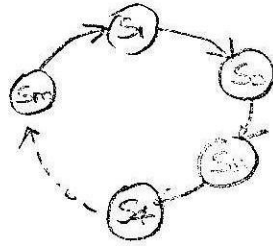


- The value on the select lines (mode control lines) of the multiplexer, decide the various operations of the shift register like the hold, shift right, shift left and parallel load.
- Each of these operations occurs during the positive edge of the clock since it is constructed using positive edge triggered D-flip flop.
- **CLEAR**: This is an active low signal, when a logic-0 is applied; it clears the contents of the register asynchronously (independent of the clock).
- When **$S_1S_0 = 00$** , I_0 of the MUX is selected. According to mode selection table, the operation that is performed is the **hold operation**.
The Q output of each flip flop gets connected to the D inputs and upon the application of the clock pulse; the data appears at the output lines.
- When **$S_1S_0 = 01$** , I_1 of the MUX is selected. According to mode selection table, the operation that is performed is the **shift right operation**.
The input to the leftmost D-flip flop is applied at the serial input for shift right line(SIR).
The input to the 2nd D-flip flop is the output of the 1st D-flip flop, hence I_1 of the second MUX is connected to Q_A . Similarly output of the 2nd D-flip flop is the input to the 3rd D-flip flop and output of the 3rd D-flip flop is the input to the 4th D-flip flop.
During the positive-edge of the clock, the contents of the register are shifted one position to the right.
- When **$S_1S_0 = 10$** , I_2 of the MUX is selected. According to mode selection table, **shift right operation** is performed.
The serial input for the left shift operation is applied to the rightmost flip flop at serial input for left shift operation (SIL).
During the positive-edge of the clock, the contents of the register are shifted one position to the left.
- When **$S_1S_0 = 11$** , I_3 of the MUX is selected. **Parallel load** is the operation performed. The parallel data inputs are applied at I_3 input of each MUX, which appears at the D input of each flip flop respectively.

COUNTERS

- Counter is a collection of flip flops configured to produce a specified output pattern sequence; hence it is also called as a **pattern generator**.
Each output sequence stored in the flip flop is referred to as the state of the counter.
- The total number of states is called as its **modulus**. Thus if a counter has 'm' states, it is called as a modulus-m-counter or **mod-m- counter**.

- A counter that counts from 0000 to 1001 has 10 states and thus is called as Mod-10-counter. State diagram is used to describe the operation of a counter. State diagram for a mod-m- counter is:

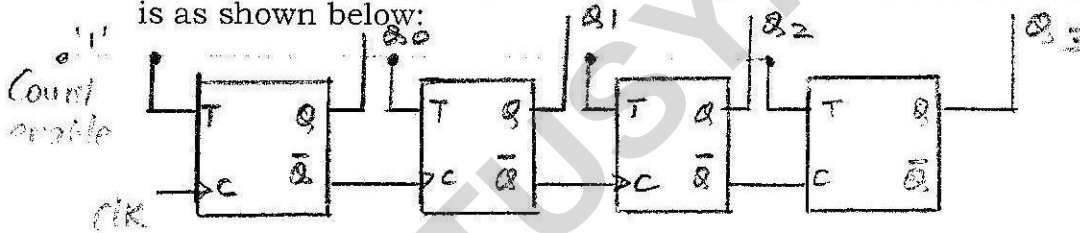


BINARY COUNTER:

- Binary counters output a binary number sequence.
- A cascade of 'n' flip flops can be used to configure a counter up to a modulus of 2^n . Thus a cascade of 4 flip flops can be used to configure a counter up to modulus 16.
- A binary up counter counts from $0_{(10)}$ to $(2^n - 1)_{(10)}$. Once the terminal count (max count value) is reached, the counting sequence is repeated.
- A binary down counter counts from $(2^n - 1)_{(10)}$ to $0_{(10)}$.

4-BIT BINARY RIPPLE COUNTER(ASYNCHRONOUS COUNTERS)

- A 4-bit binary ripple up counter configured using positive edge triggered T-flip flop is as shown below:



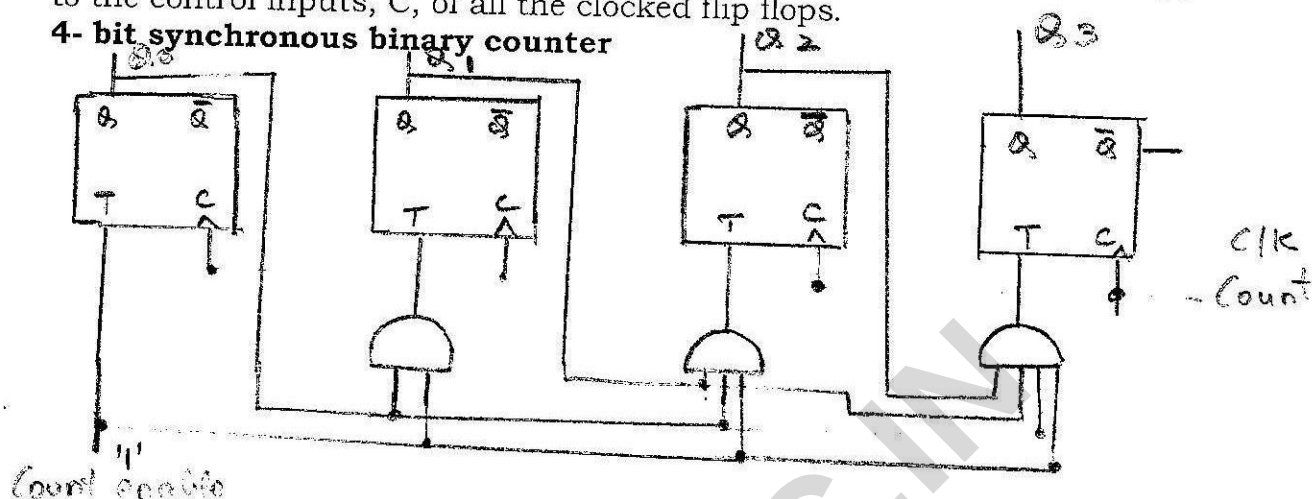
- The input to the counter is a count enable signal and a series of count pulses applied to the flip flop. Thus as long as the count enable signal is at logic-1, the Q_0 output flip flop toggles on each positive edge of the count pulse.
- The clock input of the remaining flip flops is connected to the $\bar{Q}$ output of the previous flip flop. These flip flops change their state during 0 to 1 transition of $\bar{Q}$ output, which corresponds to 1 to 0 transition of Q output.
- Since it is a 4-bit up counter, its modulus is $2^4=16$ and its counting sequence is from $0000_{(2)}$ to $1111_{(2)}$. The output of the counter appears at the Q output terminals of the 4- flip flops.
- The binary counter is known as a ripple counter since a change in state of the previous flip flop is used to toggle the flip flop next to it.
- **Ripple counters** are also known as **asynchronous counters** since clock pulse is not applied simultaneously to all the flip flops.
- There is a propagation delay between the input and output of a flip flop, the delay is maximum when all the flip flops need to toggle from 1111 to 0000 state. Thus for an n-stage binary ripple counter, the worst-case settling time is $n * t_{pd}$ where t_{pd} - propagation time delay associated with each flip flop.

Counting sequence, Timing diagram (Refer class notes)

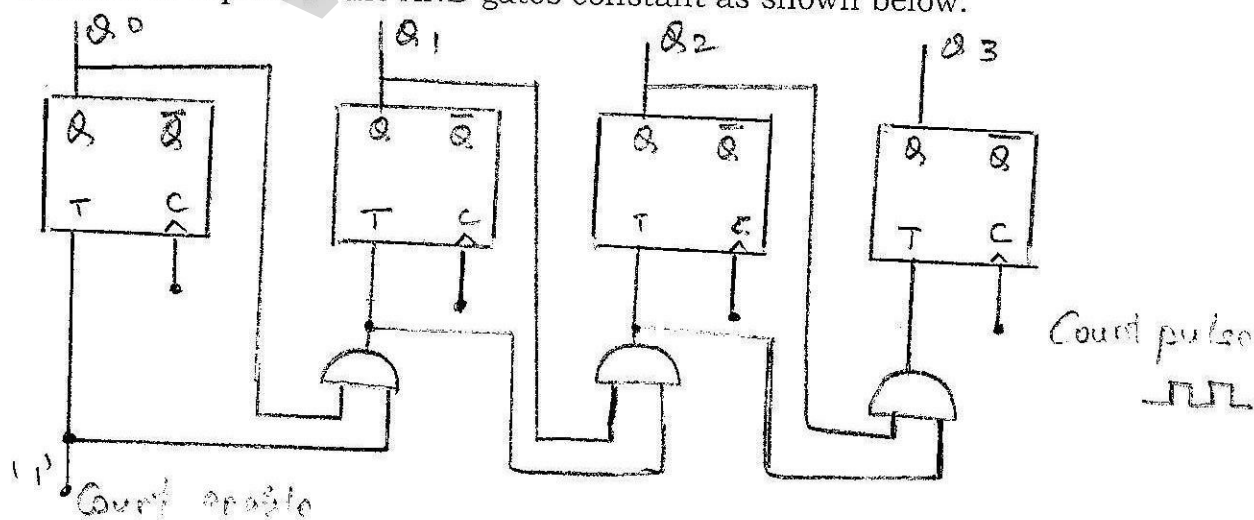
SYNCHRONOUS BINARY COUNTER

- The settling time problem associated with ripple counters is overcome in synchronous counter. In synchronous counters, the count pulses are applied directly to the control inputs, C, of all the clocked flip flops.

4-bit synchronous binary counter



- Observing the counting sequence, we find that Q_0 toggles with every clock pulse. The other outputs toggle whenever all the lower order flip flops outputs are at 1. Thus all the lower order flip flop's output can be **ANDed** to enable the toggle of a given flip flop.
- When count enable line is at logic '1', the AND gates outputs will be '1' only when all the previous flip flops outputs are at 1. This output will in turn be fed to the T-input.
- The flip flops whose T-input is at logic 1, toggles at the next clock pulse.
- The main drawback of this counter is :As the number of stages increase the number of inputs to the AND gate also increases.
- Thus making use of the fact that ANDed outputs of all previous flip flops are available at the output of each AND gate, the gating can be modified to keep the number of inputs to the AND gates constant as shown below.



4-BIT SYNCHRONOUS COUNTER WITH PARALLEL LOAD FACILITY

The logic symbol of a 4-bit synchronous binary counter with parallel load facility is as shown:

- The C_0 is the carry output which goes high during 1111 to 0000 transition.
- A logic-1 at the load input would load the count D_0, D_1, D_2, D_3 into the counter.
- A logic-1 at the count input would enable Up counting.
- The C_0 output is used to cascade several 4-bit Counters to form higher bit counter.

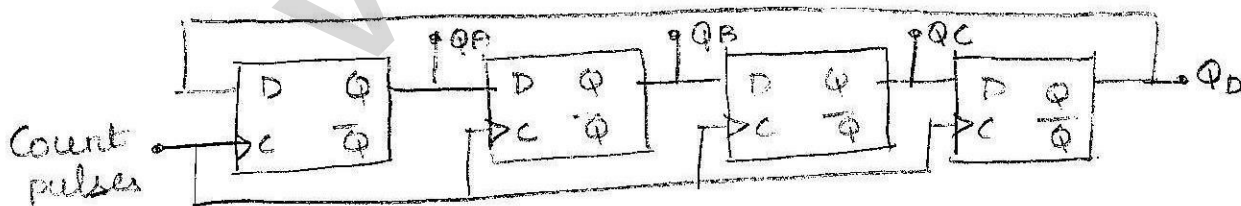
ms.

COUNTERS BASED ON SHIFT REGISTER

- The overall operation of a digital system is divided into a sequence of time-periods.
- There are two counters based on shift register.
 1. Ring counter
 2. Johnson counter

RING COUNTER

- A ring counter is a circular shift register which is initialized such that only one of its flip flop is in the 1-state; while the others are in their 0-state.
- Upon the occurrence of each count pulse, the 1 is shifted around the register.
- A ring counter consisting of 'n' flip flops has only 'n' states.
- A mod-4 ring counter configured using a circular shift register is as shown:



Initializing the counter to an initial value $Q_A Q_B Q_C Q_D = 1000$

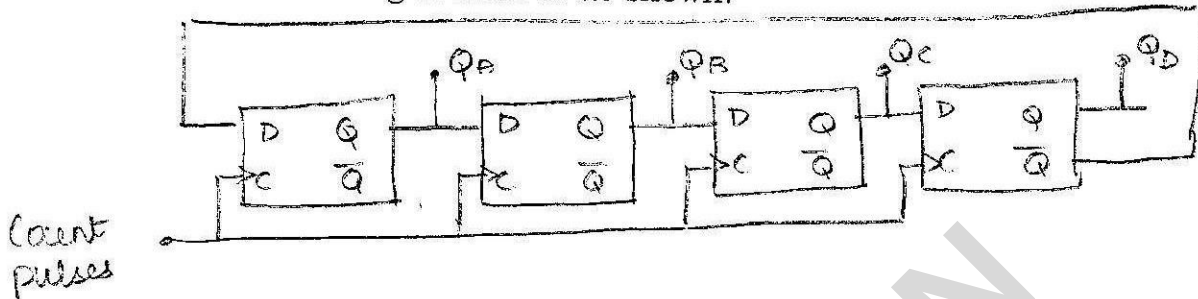
Q_A	Q_B	Q_C	Q_D
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

Repeat

Thus it can be observed that a ring counter produces a decoded output.

JOHNSON COUNTER: or TWISTED RING COUNTER or SWITCH TAIL COUNTER

- A variation of the ring counter is the switch-tail counter as shown, here the $\bar{Q}$ output of the last flip flop is provided as input to the first flip flop.
- An n-stage Johnson counter has $2n$ states.
- A two-input AND gate is sufficient to identify the decoder logic states.
- A mod-8 twisted ring counter is as shown:



Count pulses

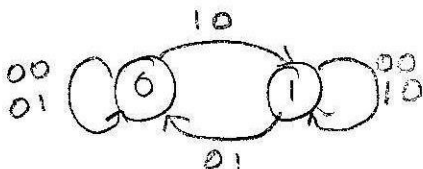
Sequence of mod-8 Johnson counter:

QA	QB	QC	QD
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1
0	0	0	0

EXCITATION TABLE OF VARIOUS FLIP FLOPS:

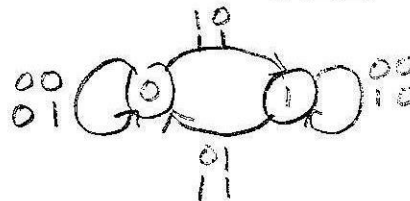
① SR Flipflop

Q	Q ⁺	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



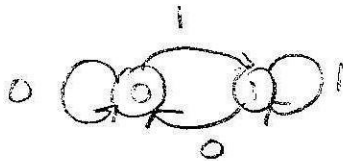
② JK Flipflop

Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



③ D Flip Flop

Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1



④ T Flip Flop

Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0



DESIGN OF SYNCHRONOUS COUNTER:

General procedure:

1. Obtain the state diagram from the given counting sequence.
2. Determine the number of flip flops.
3. Write the excitation table of the flip flop.
4. Write the excitation table of the mod-m counter.
5. Use K-map to find the expression for corresponding inputs to flip flops.
6. Draw the counter circuit by using flip flops and required gates from the above obtained Boolean expression.